FILE VERSIONING AND RECOVERY SYSTEM

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| KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- FileGuard helps users manage files, view previous versions, compare changes, and restore lost or modified content. It uses microservices, REST APIs, a database, and physical file storage.

Problem Statement & Objectives

Problem: Managing file changes and recovering previous versions can be difficult when files are accidentally modified or deleted.

- Objective 1: Maintain multiple versions of files.
- Objective 2: Compare and restore previous versions.
- Objective 3: Track file activities and protect stored versions.

Existing vs Proposed System

Feature	Existing System	Proposed System
[File Versioning]	[Limited or manual version tracking]	[Multiple versions stored automatically]
[File Recovery]	[Difficult to recover older files]	[Easy restore of previous versions]
[Activity Tracking]	[Limited activity tracking]	[Tracks file changes and user activities]
[Limitation / Gap]	[Risk of losing previous file versions]	[Secure version storage and easy recovery]

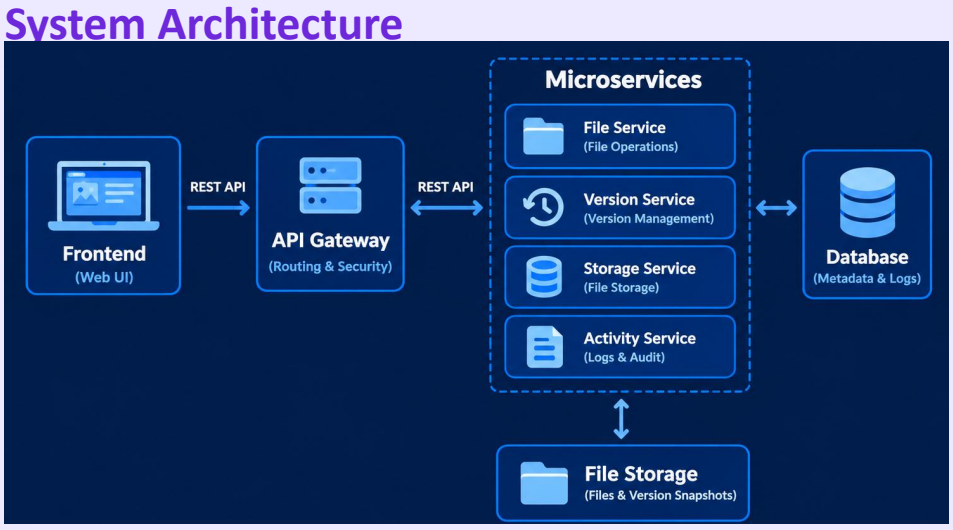
Feasibility Study

- Technical Feasibility: Uses Java, Spring Boot, REST APIs, MySQL, and VS Code.
- Economic Feasibility: Uses open-source tools, reducing development costs.
- Operational Feasibility: Provides a simple interface for file management, version tracking, and recovery.
- Schedule Feasibility: Can be developed and tested within the semester timeline.

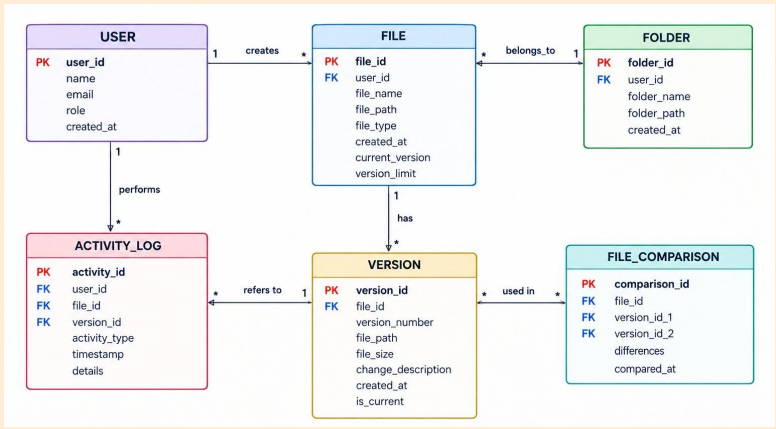
Tools & Technologies

- Front-End: HTML, CSS, JavaScript
- Back-End: Java, Spring Boot, REST APIs
- Database: MySQL
- Architecture: Microservices
- Tools: VS Code, MySQL Workbench, Maven

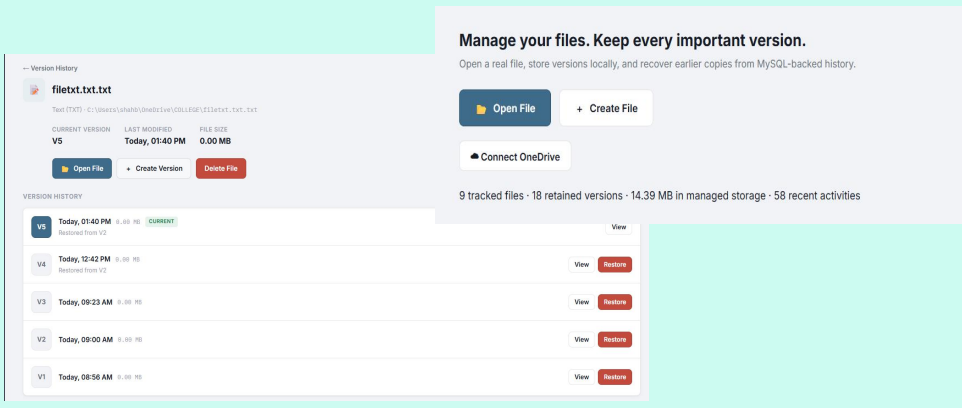
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Results & Testing

- Key Outcomes: File versioning, history viewing, comparison, and recovery.
- Testing: CRUD operations, input validation, API connectivity, and version restoration are tested to verify system reliability.

Conclusion

- Achievements: Developed a file versioning system to view, compare, and restore previous file versions.
- Objective Fulfilment: Supports reliable file management, version tracking, and recovery through microservices and database integration.objectives]

Future Scope

- Enhancement 1: Add cloud storage and automatic backups.
- Enhancement 2: Improve access control and file recovery options.